

WALMART SALES & INVENTORY ANALYSIS

SQL-Based Business Analysis and Power BI Dashboard

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1. Executive Summary

This project focuses on analysing Walmart sales data to understand the major patterns in demand, differences in store and category performance, and areas that may require attention from an inventory and demand-management perspective. The dataset covers the period from **29 January 2011 to 24 April 2016**, providing **1,913 days of sales data** for the analysis.

The main purpose of the project was not just to look at total sales, but to understand **what is driving demand, how demand changes over time, and where the business may face operational challenges**. To achieve this, I used SQL to perform different types of analysis at the product, store, category, and time levels. The results from these analyses were then used to build an interactive Power BI dashboard.

The SQL analysis covered several business questions. I first examined the overall demand trend to understand whether demand was increasing or decreasing over time. Store-level analysis was then used to identify which stores generated the highest demand and which stores showed stronger or weaker year-over-year growth. I also compared demand across product categories and examined how category demand changed over the years. In addition, the analysis looked at whether demand differed between holidays and regular days.

A major part of the project focused on understanding **product-level demand concentration**. The analysis showed that demand is not evenly distributed across all products. **The top 10% of products account for 51.14% of total demand**, while approximately 38% of products account for around 80% of total demand. This indicates that a relatively small group of products contributes a significant portion of overall demand.

The category-level analysis showed an even clearer concentration. **The FOODS category accounts for 68.63% of total unit demand**, compared with **22.04% for HOUSEHOLD** and **9.32% for HOBBIES**. This makes FOOD the dominant demand-driving category in the dataset and suggests that changes in demand within this category can have a significant effect on overall business performance.

The project also looks at inventory-related challenges from two different perspectives. First, I identified store-product combinations where demand is both relatively high and highly variable. These combinations are important because high demand combined with significant variability can make inventory planning and replenishment more difficult. For example, **FOODS_3_090** appears across several stores in the high-demand and high-variability group, with **CA_3** showing an average daily demand of 130.95 units and a standard deviation of 108.61 units.

Second, I analysed **intermittent demand**, where products do not record sales consistently throughout the dataset period. Instead of simply reporting the number of selling days, the analysis expresses this as the **percentage of days on which a product recorded sale**. For example, **FOODS_3_296** recorded sales on only about **10.04% of the 1,913 days**, indicating a highly intermittent sales pattern.

The final Power BI dashboard brings these analyses together across two pages. **Page 1 provides a business overview**, covering overall demand, store performance, category contribution, demand variability, holiday effects, and store growth. **Page 2 focuses on deeper product and inventory analysis**, including product demand concentration, high-demand variability, intermittent demand, and category trends over time.

Overall, the project combines SQL analysis with Power BI visualization to move from raw sales data to a clearer understanding of the business. The findings can help management focus attention on the products, stores, and categories that have the greatest impact on demand, while also highlighting areas where demand patterns may make inventory planning more challenging.

2. Business Problem & Objectives

2.1 Business Problem

Walmart operates across many stores and sells a wide variety of products belonging to different categories. With sales taking place across different dates, stores, and products, the resulting dataset contains a large amount of information about how demand behaves. However, simply looking at total units sold does not provide enough information to understand the underlying business situation. Two products may have similar total demand but very different sales patterns, while two stores may have similar demand levels but very different levels of variability or growth.

One of the main challenges is therefore to **identify the factors and patterns that drive overall demand**. The business needs to understand whether demand is changing over time, which stores contribute the most to total demand, and which product categories are responsible for a significant share of sales. Understanding these patterns can help provide a clearer picture of where the business is performing strongly and where changes in demand may require further attention.

Another important issue is **that demand is not necessarily distributed evenly across products**. A small number of products may account for a large proportion of total demand, making them more important from a business and inventory perspective. Identifying this concentration can help distinguish products that have a major impact on overall demand from products that contribute relatively little. At the same time, category-level analysis can show whether the business is heavily dependent on categories.

The project also considers the fact that **demand can behave differently across stores and products**. High demand is generally positive, but when high demand is accompanied by substantial variability, predicting future requirements and maintaining appropriate inventory levels can become more difficult. On the other hand, some products may have intermittent demand, meaning they record sales on only a small proportion of the days in the dataset. These two situations represent different demand-management challenges and therefore need to be analysed separately.

Seasonality and calendar-related effects are another consideration. Demand may not remain consistent throughout the year, and customer purchasing behaviour can differ during holidays compared with regular days. Similarly, comparing stores across years can reveal whether their demand is increasing or declining. Understanding these changes is important because looking at a single overall sales figure can hide important differences in how demand is evolving.

Therefore, the business problem addressed in this project is to **transform the available Walmart sales data into meaningful information about demand, store performance, product concentration, category**

contribution, and potential inventory challenges. The analysis uses SQL to investigate these questions at different levels of detail and Power BI to present the results in an interactive and understandable format. The aim is not only to describe what happened in the data, but to identify patterns that can help management understand where demand is concentrated, how it is changing, and which areas may require greater attention.

2.2 Project Objective

The primary objective of this project is to analyse Walmart sales data and identify patterns that can provide a better understanding of **demand, store performance, product concentration, and potential inventory challenges**.

The project aims to:

1. **Understand how overall demand changes over time** and identify important demand trends.
2. **Evaluate differences in store performance and growth** to identify stores with stronger or weaker demand patterns.
3. **Understand the contribution of different product categories** and determine how category demand changes over time.
4. **Examine differences in customer demand during holidays and regular days** to identify meaningful calendar-related demand patterns.
5. **Measure product demand concentration** to determine whether a relatively small group of products drives a significant portion of total demand.
6. **Identify high-demand products and store-product combinations with substantial demand variability**, highlighting areas that may require greater attention in inventory planning.
7. **Identify products with intermittent demand** and distinguish them from products with more consistent sales activity.
8. **Develop an interactive Power BI dashboard** that communicates these findings clearly and allows users to explore the results.
9. **Translate the analysis into actionable business insights** rather than simply reporting descriptive statistics.

3. Dataset Description & Understanding

3.1 Dataset Overview

The project uses a Walmart sales database containing information about product sales across different stores and dates. The data provides a detailed view of how many units of different products were sold at individual stores over time.

The main sales data is stored in the `sales_fact` table. Each sales record contains information about the product, store, category, date, and number of units sold. The primary measure used in this project is `units_sold`, which is used to analyse demand across different levels of the business.

The sales data covers the period from **29 January 2011 to 24 April 2016**, representing **1,913 calendar days**. This period provides enough observations to examine changes in demand over time, differences between stores and products, and patterns such as holiday demand and intermittent sales.

In addition to the main sales table, the database contains a calendar table that provides additional information about dates, including the year, month, weekday, and events or holidays. This information was useful for analysing time-based demand patterns and comparing holiday demand with regular-day demand.

The database also contains a `sell_prices` table with weekly selling-price information for products across stores. This table was reviewed as part of the dataset exploration but was not included in the primary dashboard analysis, as the main focus of the project was demand behaviour, store performance, product concentration, and inventory-related patterns.

A `vw_store_growth` view is also available in the database. It contains annual store-level demand and year-over-year growth information and supports the analysis of changes in store performance over time.

3.2 Tables in the Database

TABLE/VIEW	PURPOSE
<code>Sales_fact</code>	Main sales table containing product, store, category, date, and units sold information
<code>calendar</code>	Provides date, year, month, weekday, and event/holiday information
<code>Sell_price</code>	Contains weekly selling-price information for products across stores
<code>vw_store_growth</code>	Provides annual store demand and year-over-year growth information

3.3 Important Fields

Field	Description
sale_id	Unique identifier for each sales record
sale_date	Date on which the sale was recorded
store_id	Identifier of the store
item_id	Identifier of the product
dept_id	Identifier of the product department
cat_id	Identifier of the product category
state_id	Identifier of the state associated with the store
units_sold	Number of units sold; used as the primary measure of demand
date	Calendar date used for time-based analysis
year	Year associated with the calendar date
month	Month associated with the calendar date
weekday	Day of the week
event_name_1	Name of the primary event or holiday
event_type_1	Type of the primary event or holiday
sell_price	Weekly selling price of a product at a particular store

4. Data Preparation & Data Modelling

4.1 Data Handling Challenge

A key challenge during the project was handling the size of the raw datasets. The initial plan was to import the raw data directly into Power BI. However, one dataset contained approximately **65 million rows**, while another contained around **6 million rows**. Loading and combining these datasets in Power BI led to storage and performance limitations, making this approach impractical.

4.2 Approach Used

To overcome this limitation, the heavy data processing was moved to **SQL Server**. Instead of bringing the complete raw datasets into Power BI, SQL was used to perform the required aggregations, calculations, and business analyses first.

This produced smaller, analysis-ready datasets containing only the information required for each business question. These results were then imported into Power BI for visualization.

4.3 Final Workflow

The resulting workflow was:

Raw Data → SQL Server → Analytical Tables/Views → Power BI Dashboard

SQL Server handled the data processing and analytical calculations, while Power BI was used mainly for visualization and presenting the findings.

This approach reduced the amount of data that Power BI had to handle and made it possible to build the dashboard without loading the complete raw datasets.

5. SQL Analysis & Business Questions

SQL was used to answer specific business questions from the sales data. The analysis was performed at different levels such as year, store, category, product, and store-product combination. The results were then used in the Power BI dashboard.

NOTE: The complete SQL queries used for these analyses are provided in [Appendix A](#).

5.1 Overall Demand Trend

Business Question:

How does overall demand change over time?

Approach:

Total units_sold was grouped by year to see how demand changed from 2011 to 2016.

Key Result:

The analysis shows the overall movement of demand across the dataset period. However, **2016 contains data only up to 24 April**, so it represents a partial year.

Business Relevance:

This gives a quick view of the overall demand pattern and provides a baseline for the store, category, and product-level analysis.

Limitation:

2016 should not be compared with complete years as a full-year figure. Also, the analysis measures **units sold**, not revenue or profit.

Power BI Visual: Yearly Demand Trend

5.2 Store Performance & YoY Growth

Business Question:

How do stores differ in demand and growth?

Approach:

Annual demand was calculated for each store. **LAG() was then used** to compare each year's demand with the previous year and calculate YoY growth.

Key Results:

- **WI_2** had very strong early growth: **79.91% in 2012** and **37.57% in 2013**.
- **WI_1** grew by **40.77% in 2013** and continued growing in 2014 and 2015.
- **TX_3** showed steady positive growth in both 2014 (**8.77%**) and 2015 (**8.66%**).
- **WI_3** declined in 2013 and 2014 but recovered with **13.17% growth in 2015**.

The large negative growth values shown for **2016 are mainly due to the year being incomplete**, rather than evidence of a full-year decline.

Business Relevance:

This analysis shows that store performance is not the same everywhere. It helps distinguish stores with **high demand** from stores with **strong growth**.

Limitation:

YoY growth should be considered alongside total demand because a smaller store can have a high growth rate but still generate less demand than a larger store.

Power BI Visuals: Store Performance by Demand and Store YoY Growth

5.3 Category Demand Contribution & Trend

Business Question:

Which categories contribute most to demand, and does their contribution change over time?

Approach:

Total units sold were grouped by category to calculate each category's contribution to overall demand. Category demand was also analysed across years to observe changes in category share.

KEY RESULTS:

Category	Units Sold	Demand Share
FOODS	45,089,939	68.63%
HOUSEHOLD	14,480,670	22.04%
HOBBIES	6,124,800	9.32%

FOODS is clearly the dominant category, accounting for more than two-thirds of total unit demand.

Business Relevance:

The category mix shows where most of the demand is concentrated. Since FOODS contributes the largest share, changes in its demand can have a noticeable effect on overall demand.

The category trend also helps determine whether this contribution remains relatively stable or changes over time.

Limitation:

The analysis measures unit demand and does not indicate which category generates the highest revenue or profit.

Power BI Visuals: Overall Category Demand Share and Category Demand Share Trend

5.4 Holiday vs Regular Demand

Business Question:

Does customer demand differ between holidays and regular days?

Approach:

Sales dates were matched with the calendar information to separate holiday/event days from regular days. Total demand was then compared between the two groups.

Key Result:

The analysis provides a direct comparison of demand behaviour during holiday/event periods versus regular trading days.

Business Relevance:

Understanding calendar-related demand changes can help the business plan inventory and replenishment around periods where customer purchasing behaviour may differ.

Limitation:

A holiday comparison shows an overall difference between the two groups, but it does not by itself explain which specific products or stores are responsible for the change.

Power BI Visual: Holiday vs Regular Demand

5.5 Product Demand Concentration

Business Question:

Is demand evenly distributed across products, or is it concentrated among a smaller group?

Approach:

Products were ranked from highest to lowest total demand. Their individual demand share and cumulative demand share were then calculated.

Key Results:

- The **top 10% of products account for 51.14% of total demand.**
- Approximately **38% of products account for around 80% of total demand.**

This shows that demand is **not evenly distributed** across the product portfolio.

Business Relevance:

A relatively small group of products has a major influence on total demand. These products can therefore receive greater attention when prioritising inventory and replenishment decisions.

Limitation:

The analysis identifies demand concentration but does not consider profit margins, product costs, or actual inventory availability.

Power BI Visual: Product Demand Concentration

5.6 High-Demand Products with Highest Demand Variability

Business Question:

Which store-product combinations have both relatively high demand and high demand variability?

Approach:

The analysis considered all **30,490 store-product combinations**.

- Overall average daily demand: **1.13 units**
- Combinations above this average: **7,194**
- Average standard deviation among these high-demand combinations: **3.84 units**
- High-demand combinations also above this variability level: **2,116**
- The final visual shows the **top 15** combinations by demand variability.

One clear example is **FOODS_3_090 at CA_3**, with:

- Average daily demand: **130.95 units**
- Standard deviation: **108.61 units**

FOODS_3_090 also appears across several stores in the high-demand/high-variability group.

Business Relevance:

High demand combined with high variability can make demand planning and replenishment more difficult. These combinations therefore deserve closer operational attention.

Limitation:

This is a **demand-based risk indicator**, not evidence of actual stockouts or inventory shortages. The analysis does not contain actual inventory or stockout information.

Power BI Visual: High-Demand Products with Highest Demand Variability

5.7 Intermittent Demand

Business Question:

Which products have sales on only a small proportion of the available days?

Approach:

For each product, the number of days with positive sales was counted. This was then converted into the **percentage of days with sales** over the 1,913-day dataset period.

Key Result:

For example, **FOODS_3_296** recorded sales on only **192 out of 1,913 days**, meaning it had sales on approximately **10.04% of the available days**.

The analysis therefore identifies products with very infrequent sales activity.

Business Relevance:

Intermittent products behave differently from consistently selling products. Identifying them can help distinguish products that may require different approaches to inventory planning and replenishment.

Limitation:

The measure captures **how frequently sales occurred**, not the length of the gaps between sales. Therefore, it should not be interpreted as a direct measure of the number of days between purchases.

Power BI Visual: Most Intermittent Products

5.8 Demand Volume vs Variability by Store

Business Question:

Which stores combine relatively high demand with greater demand variability?

Approach:

Average daily demand and demand standard deviation were calculated for each store. These two measures were plotted against each other using a scatter plot.

Key Result:

The visual shows that stores differ not only in their demand volume but also in how variable that demand is.

Business Relevance:

Looking at demand volume and variability together gives a more useful picture than looking at either measure alone. A store with high demand and high variability may require more attention from a demand-planning perspective.

Limitation:

The analysis identifies differences in demand behaviour but does not explain the underlying reasons for those differences. Factors such as store characteristics, local customer behaviour, or promotions are not available in the current analysis.

Power BI Visual: Demand Volume vs Variability by Store

6. Power BI Dashboard Development

6.1 Dashboard Overview

The SQL analysis produced several analysis-ready datasets, which were then used to develop an interactive dashboard in Power BI. The main aim of the dashboard was to present the analysis in a simple and business-friendly way, allowing the user to understand overall demand as well as deeper product and inventory-related patterns.

The dashboard contains **10 visuals divided across two pages**. Each visual was selected to answer a specific business question, while avoiding unnecessary duplication of information.

6.2 Page 1 – Business Overview

The first page provides a high-level view of demand and store performance.

S. No.	Visual Name	Purpose
1	Yearly Demand Trend	Shows how overall demand changes over time
2	Store Performance by Demand	Compares demand across stores
3	Overall Category Demand Share	Shows the contribution of each category to total demand
4	Demand Volume vs Variability by Store	Compares store demand volume with demand variability
5	Holiday vs Regular Demand	Compares demand during holidays and regular days
6	Store YoY Growth	Shows year-over-year demand growth across stores

The first page is designed to give management a quick understanding of **what is happening across the business** before moving into more detailed product-level analysis.

6.3 Page 2 – Strategic & Operational Insights

The second page focuses on product-level patterns and areas that may require greater attention from an inventory and demand-management perspective.

S. No.	Visual Name	Purpose
1	Product Demand Concentration	Shows how demand is distributed across products
2	High-Demand Products with Highest Demand Variability	Identifies high-demand store-product combinations with substantial demand variability
3	Most Intermittent Products	Identifies products with infrequent sales activity
4	Category Demand Share Trend	Shows how category contribution changes over time

This page moves from general business performance towards **more specific demand and inventory-related insights**.

6.4 Interactivity and User Controls

Interactive features were added where they improved the interpretation of the analysis.

The **Store YoY Growth** visual includes a **year slicer**, allowing users to select a particular year and compare store-level growth for that period.

Tooltips were also used where appropriate to provide additional information without overcrowding the main visuals.

The dashboard was intentionally kept limited to **10 visuals** so that the information remains readable and each visual has a clear purpose.

6.5 Dashboard Design Approach

The dashboard follows a simple progression:

[Overall Demand](#) → [Store Performance](#) → [Category Behaviour](#) → [Product Concentration](#) → [Demand Variability](#) → [Intermittent Demand](#)

This structure allows the user to first understand the overall business situation and then move towards the product-level patterns that may have greater operational importance.

The dashboard focuses on **clarity and business interpretation rather than the number of charts**, with each visual linked to one of the questions addressed through the SQL analysis.

7. Project Limitations

Like any analysis based on historical sales data, this project has some limitations that should be considered when interpreting the results.

7.1 Partial 2016 Data

The dataset ends on **24 April 2016**, so 2016 does not represent a complete year. Therefore, the 2016 values, particularly the YoY growth figures, should not be interpreted as full-year performance.

7.2 Demand Measured Using Units Sold

The primary measure used in the project is **units sold**. The analysis therefore focuses on demand volume rather than revenue, profit, or margins.

7.3 No Actual Inventory or Stockout Data

The project identifies potential inventory-related concerns using **demand variability and intermittent demand**. However, actual inventory levels, stockouts, replenishment quantities, and lost sales are not available in the analysed data.

Therefore, the project highlights **potential inventory-planning challenges rather than confirmed stockout problems**.

7.4 Limited Explanation of Demand Drivers

The analysis identifies patterns in demand across stores, products, categories, and time. However, the available data does not provide enough information to determine the exact reasons behind every change. Factors such as local customer behaviour, promotions, or store-specific conditions are not directly analysed.

7.5 Price Data Not Used in the Primary Analysis

Weekly selling-price data was available and reviewed, but it was not included in the primary dashboard analysis. Therefore, the project does not examine how changes in selling price may have affected demand.

7.6 Power BI Data Handling

Due to the size of the raw datasets, the complete raw data was not loaded directly into Power BI. Heavy processing and aggregation were performed in SQL Server, and the resulting analysis-ready datasets were used for dashboard development. This approach was necessary to overcome the storage and performance limitations encountered during the initial Power BI import.

8. Challenges Faced and How They Were Overcome

During the project, the main challenges were related to **data size, data structure, analysis methodology, and dashboard design**. The following steps were taken to address them.

8.1 Handling Large Data Volume

Challenge:

The sales_fact table contained approximately **65 million rows**, while the sell_prices table contained around **6 million rows**. When the two tables were combined through a relationship in Power BI, the resulting data became too large to be handled efficiently with the available local storage.

How it was overcome:

The heavy data processing was moved to **SQL Server**. Required calculations and aggregations were performed there, and only smaller, analysis-ready datasets were imported into Power BI.

Outcome:

This reduced the data-processing and storage burden on Power BI and made dashboard development feasible.

8.2 Converting Raw Data into a Usable Structure

Challenge:

The original sales data was in a wide format, with dates represented across multiple columns. This structure was not convenient for time-based analysis.

How it was overcome:

The sales data was transformed into a long format, with the date and units sold represented as separate fields. This made it easier to perform time-based and other analytical operations.

Outcome:

The resulting structure was more suitable for SQL analysis and Power BI visualization.

8.3 Choosing Appropriate Measures for Demand Analysis

Challenge:

Some initial measures did not represent the business problem effectively. For example, using only the number of selling days did not provide a comparable measure of intermittency across the full observation period.

How it was overcome:

The analysis was refined to use **percentage of days with sales**, based on the 1,913-day dataset period. Similarly, high-demand and high-variability products were identified using separate demand and variability thresholds rather than relying on a single combined score.

Outcome:

The final measures provided a clearer and more meaningful representation of intermittent and variable demand.

8.4 Handling Partial-Year Data

Challenge:

The dataset ends on **24 April 2016**, making 2016 an incomplete year. This created a problem when interpreting year-over-year growth.

How it was overcome:

2016 was treated as a **partial year**, and its YoY figures were not interpreted as full-year performance. A year slicer was also provided in the dashboard to make year-specific comparisons easier.

Outcome:

This reduced the risk of drawing misleading conclusions from incomplete-year data.

8.5 Keeping the Dashboard Focused

Challenge:

The project generated several possible analyses and visualizations, but including too many visuals could make the dashboard difficult to read and reduce its business focus.

How it was overcome:

The final dashboard was limited to **10 visuals across two pages**, with each visual linked to a specific business question.

Outcome:

The dashboard remained focused while still covering the major demand, store, category, and product-level findings.

9. Conclusion

This project used **SQL Server and Power BI** to analyse Walmart sales data from January 2011 to April 2016. The analysis examined demand trends, store performance, category contribution, holiday effects, product concentration, demand variability, and intermittent demand.

The results show that demand is not evenly distributed across the business. A relatively small proportion of products contributes a large share of total demand, while the **FOODS category alone accounts for 68.63% of total unit demand**. The analysis also identified store-product combinations where high demand is accompanied by substantial variability, as well as products with highly intermittent sales activity.

Overall, the project demonstrates how large-scale sales data can be transformed into meaningful business information by combining **SQL-based analysis with Power BI visualization**. The dashboard provides a practical way to explore the findings, while the separate Business Insights & Recommendations document translates these findings into management-focused actions.